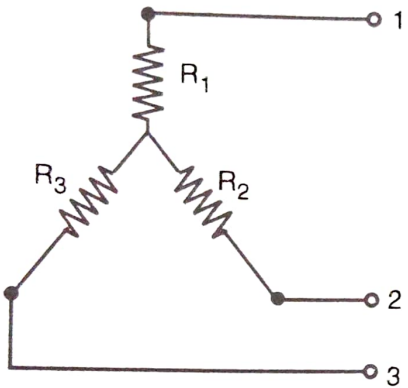
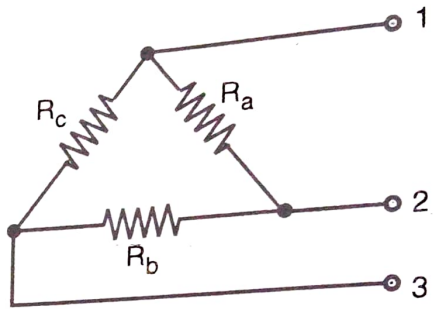
3.4. STAR-DELTA CONNECTION

When three resistors R_1, R_2 and R_3 are connected as shown in Fig. 3.16 (a), then this connection is known as the star connection or Y-connection. Here one ends of three



(a) Star Connection



(b) Delta Connection

Fig. 3.16.

resistors are connected together to form neutral or star point while other ends are free.

Similarly, if three resistors are connected to form a closed path as shown in Fig. 3.16 (*b*), then this connection is known as *the delta* or *D-connection.*

Star-delta conversion. The star connection can be converted into delta connection and vice-versa. First we convert the star connection to delta connection (See Fig. 3.17).

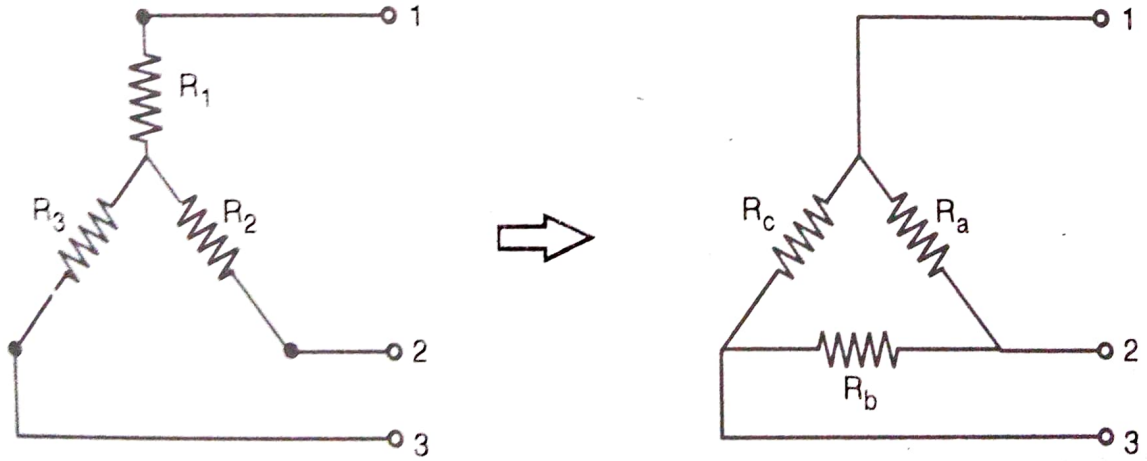
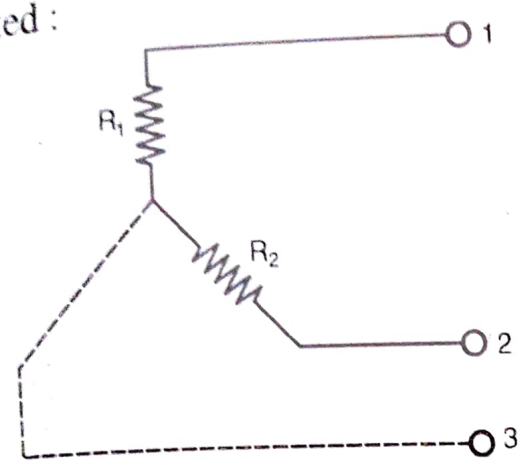


Fig. 3.17.

$R_1, R_2, R_3 \rightarrow$ known resistance

$R_a, R_b, R_c \rightarrow$ unknown resistance

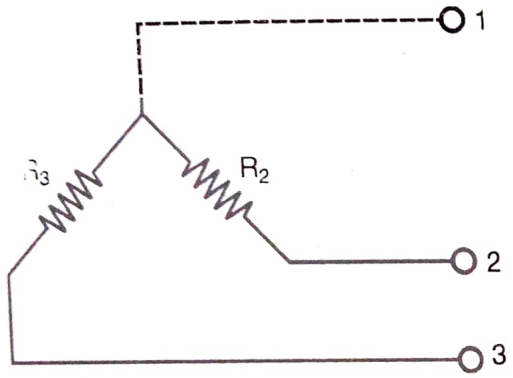
From the star connection. Resistance between terminals 1 and 2 when 3 is open circuited :



$$\Rightarrow R_{12} = R_1 + R_2 \qquad \ldots (3.28)$$

Fig. 3.18.

Similarly, resistance between terminals 2 and 3 when terminal one is open circuited.



$$\Rightarrow R_{23} = R_2 + R_3 \qquad \ldots (3.29)$$

Fig. 3.19.

In the same way resistance between terminal, 1 and 3 is

$$R_{13} = R_1 + R_3. \qquad \ldots (3.30)$$

From the delta connection. Resistance between terminals 1 and 2 when 3 is o circuited.

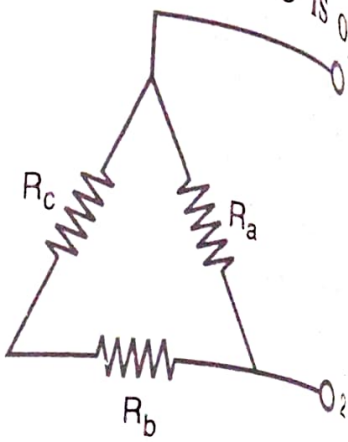


Fig. 3.20.

R_c and R_b are in series. This combination is in parallel to R_a.

$$R_{12} = \frac{R_a (R_b + R_c)}{R_a + R_b + R_c} \qquad ...(3.31)$$

Similarly, the resistances between terminals 2 and 3 and 1 and 3 are given by :

$$R_{23} = \frac{(R_a + R_c) R_b}{R_a + R_b + R_c} \qquad ...(3.32$$

$$R_{13} = \frac{(R_a + R_b) R_c}{R_a + R_b + R_c} \qquad ...(3.33$$

Now compare the results of star and delta connections :

$$R_1 + R_2 = \frac{R_a (R_b + R_c)}{R_a + R_b + R_c} \qquad ...[3.34 (i)]$$

$$R_2 + R_3 = \frac{R_b (R_a + R_c)}{R_a + R_b + R_c} \qquad ...[3.34 (ii)]$$

$$R_1 + R_3 = \frac{R_c (R_a + R_b)}{R_a + R_b + R_c} \qquad ...[3.34 (iii)]$$

After solving the equations 3.34 (i), (ii), (iii) we can determine three unknown resistances R_a, R_b and R_c in terms of R_1, R_2 and R_3.

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} \qquad ...(3.35)$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} \qquad ...(3.36)$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} \qquad ...(3.37)$$

The equations 3.39, 3.40 and 3.41 are the values of resistances in delta connection in terms of the star resistances.

terms of the star resistances.

Delta to star conversion. Now we have to find R_1, R_2 and R_3 in terms of R_a, R_b and R_c.

R_a, R_b, R_c – Known resistances

R_1, R_2, R_3 – Unknown resistances.

Again solving the equations [3.38 (*i*) (*ii*), (*iii*)] we can determine the resistances of star connection *i.e.*, R_1, R_2 and R_3.

$$R_1 = \frac{R_a R_c}{R_a + R_b + R_c} \qquad \ldots(3.38)$$

$$R_2 = \frac{R_a R_b}{R_a + R_b + R_c} \qquad \ldots(3.39)$$

$$R_3 = \frac{R_b R_c}{R_a + R_b + R_c} . \qquad \ldots(3.40)$$

Example 2.9. *Obtain the equivalent inductance at terminal AB in circuit shown in Fig. 2.49.*

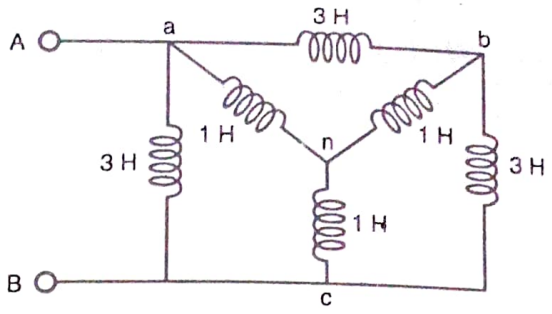


Fig. 2.49.

Solution. Inductances also follow the same rule as the resistances for series, parallel and star-delta connections. First we convert the star connection *a b c n* to corresponding delta connection.

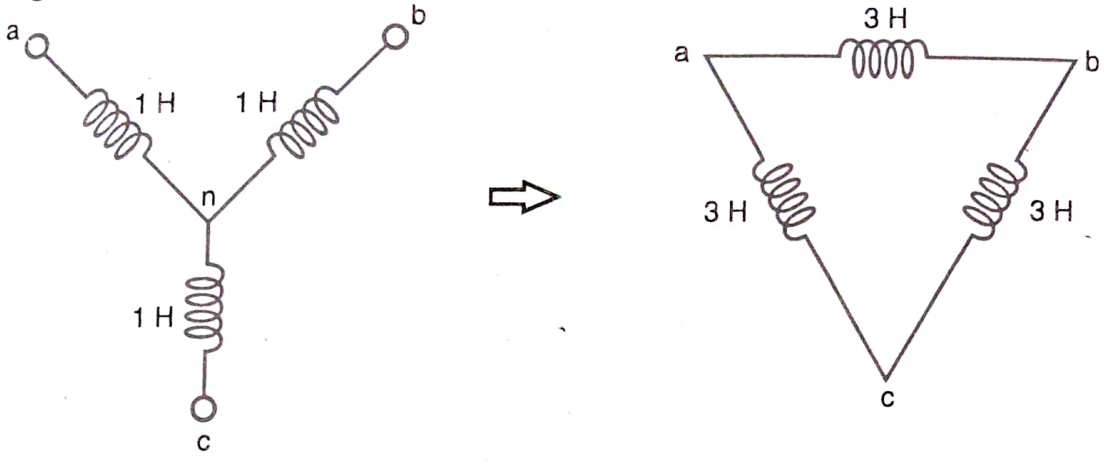


Fig. 2.50.

For delta connection,

$$L_{ab} = \frac{L_{an}\,L_{bn} + L_{bn}\,L_{cn} + L_{cn}\,L_{an}}{L_{cn}}$$

$$= \frac{1+1+1}{1} = 3\text{H}$$

Similarly, $L_{bc} = 3\text{H}$ and $L_{ac} = 3\text{H}$

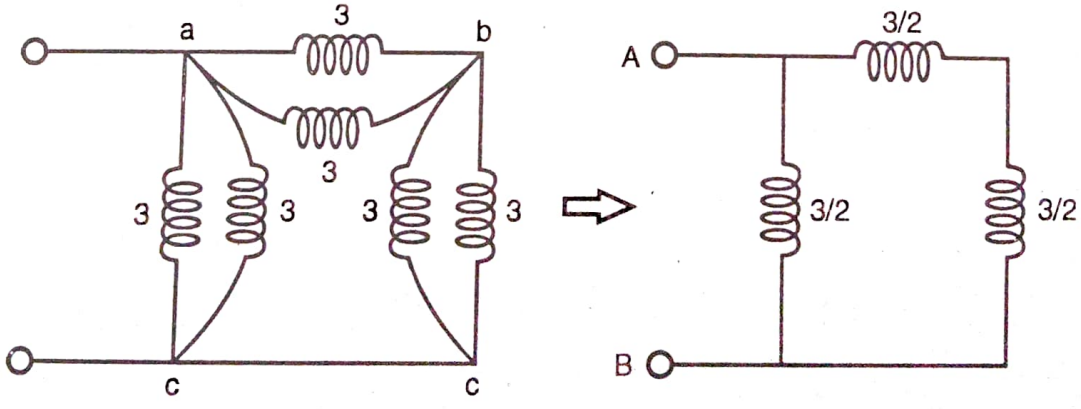


Fig. 2.51.

So the circuit of Fig. 2.49, is redrawn as shown in Fig. 2.51, 2.52.

Example 2.7. *Find R_{eq} in the network shown in Fig. 2.42.*

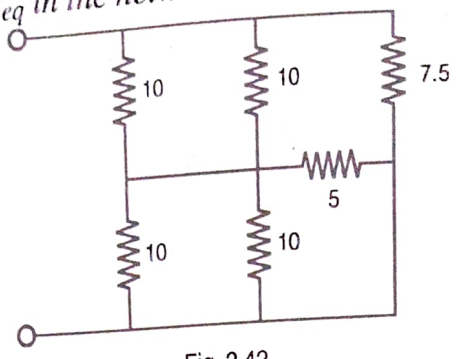


Fig. 2.42.

Solution. Solving the parallel branches. The given circuit become,

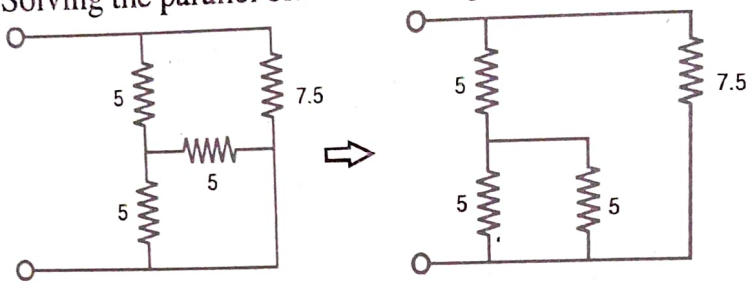


Fig. 2.43.

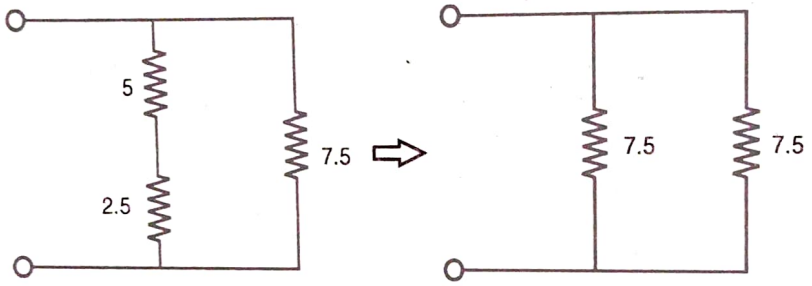


Fig. 2.44.

$$R_{eq} = 7.5 \parallel 7.5 = \frac{7.5 \times 7.5}{7.5 + 7.5} = 3.75\ \Omega$$ **Ans.**

Example 2.8. *Find R_{eq} in Fig. 2.45*

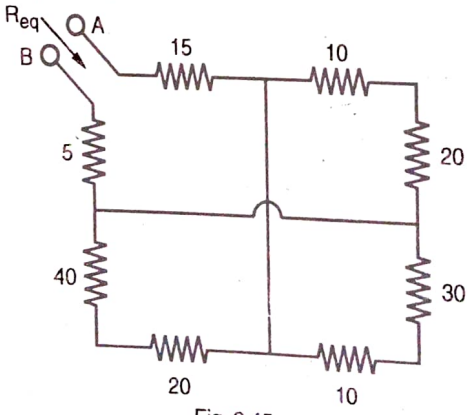


Fig. 2.45.

Solution. The given circuit may be drawn as :

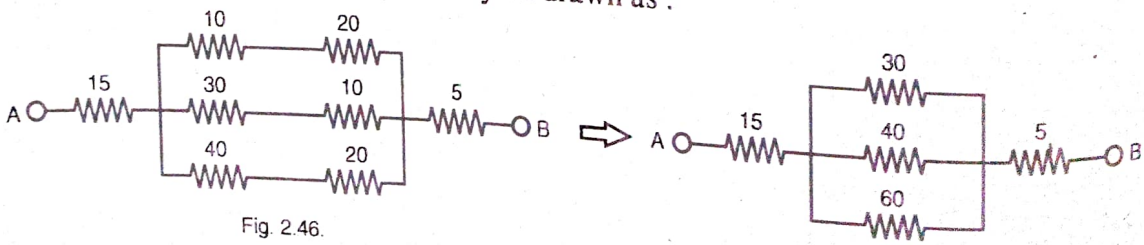


Fig. 2.46.

Fig. 2.47.